

Lecture 13

Tues 7/21 PM

outline/topics

- Particle source fundamentals
- Electron emission
- Solenoid focusing

~~Ion sources~~

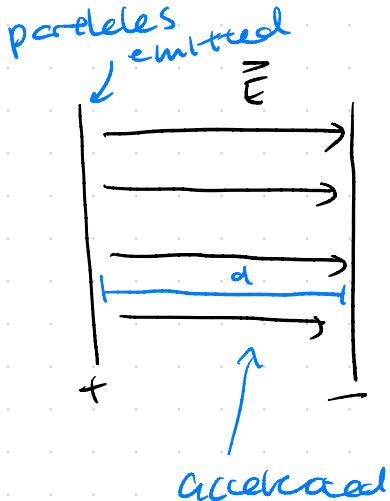
post-poned to Thurs

Fundamentals

- Basics we've been discussing still apply, but at a lower energy
- Both $\vec{E}$ and $\vec{B}$ play important roles

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

- Behavior is more transient
- * Almost nothing in H.W. or S.Y. Lee
Source for those interested:
Martin Reiser "Theory + Design of
Charged Particle Beams"
- Basic picture: "1D diode"



(direction
shown for
positive
charges)

• Same idea as previous acceleration discussion

• Field related to the scalar potential

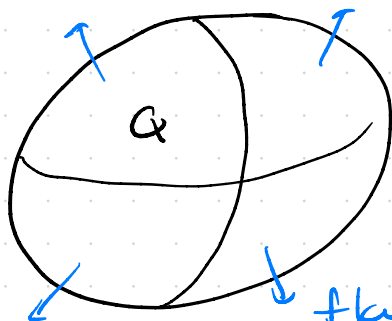
$$\vec{E} = -\vec{\nabla}\phi \quad \text{or} \quad \phi = -\int \vec{E} \cdot d\vec{r}$$

• Maxwell's equations: now considering free charges + currents

$$\vec{\nabla} \cdot \vec{D} = \rho \quad \text{and} \quad \vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} + \vec{J}$$

$$\text{with } \vec{D} = \epsilon \vec{E}, \quad \vec{B} = \mu \vec{H}$$

• Continuity equation: conservation of charge



Sphere with charge density

$$\rho = \frac{Q_{\text{enc}}}{\text{volume}}$$

flow of charges out

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0$$

change in ρ must be due to $\vec{\nabla} \cdot \vec{J}$

e.g. $\frac{\partial \rho}{\partial t} < 0$, $\vec{\nabla} \cdot \vec{J} > 0$ are not allowed unless

Can derive from Maxwell's equations
(simple proof)

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0 \text{ always}$$

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = \vec{\nabla} \cdot \left(\frac{\partial \vec{D}}{\partial t} + \vec{J} \right) = 0$$

$$\frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0 \Rightarrow \left(\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0 \right)$$

$\vec{\nabla} \cdot \vec{D} = \rho$ Gauss's Law

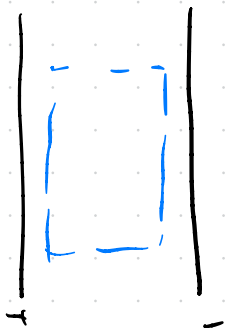
Relate to potential:

$$\vec{\nabla} \cdot \vec{D} = \rho \Rightarrow \vec{\nabla} \cdot \vec{E} = \rho / \epsilon$$

$$\text{with } \vec{E} = -\vec{\nabla} \phi \Rightarrow \vec{\nabla} \cdot \vec{\nabla} \phi = -\rho / \epsilon$$

$$\boxed{\nabla^2 \phi = -\rho / \epsilon} \quad \text{Poisson's equation}$$

- Consider surface in front of
an emission surface



Right now: not
yet discussing
How particles are
emitted (more soon)

- Two cases in this 1D example
 - without space charge, i.e. neglecting effects from particles which have already been emitted:

$$\rho = 0 \Rightarrow \nabla^2 \phi = 0, \text{ solution}$$

$$\phi(z) = \frac{V_0}{d} z$$

$\Rightarrow$ uniform $\vec{E}$ we expect:

$$E_z = -\frac{d\phi}{dz} = -\frac{V_0}{d} \quad \begin{array}{l} V_0 \text{ gap} \\ \text{voltage} \end{array}$$

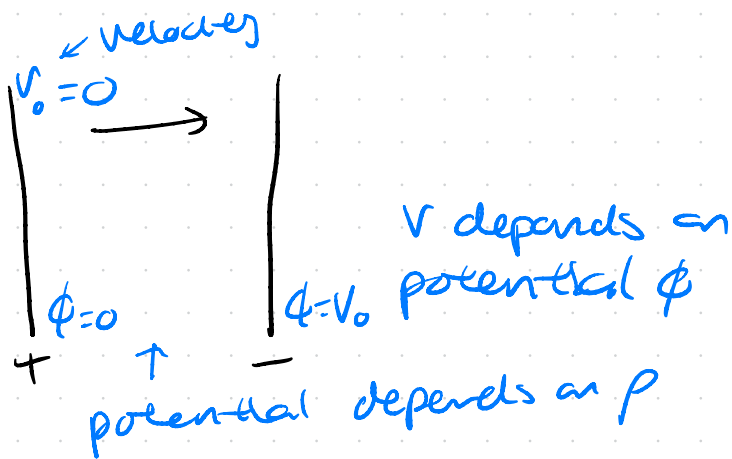
- WITH space charge, more realistic emitted charges change ϕ

$$\rho \neq 0, \quad \frac{d^2\phi}{dz^2} = \frac{\rho}{\epsilon} \quad (1D \text{ case})$$

Need to consider the electrostatic potential from the beam itself

- Want steady state solution, assume "laminar" flow where current density is uniform + particle trajectories do not cross $\Rightarrow \vec{\nabla} \cdot \vec{J} = 0$

$$\Rightarrow J = \rho v \text{ constant}$$



- Equation of motion $\frac{m}{2} \dot{z}^2 = q \phi(z)$

↳ From continuity: $J_z = \rho \dot{z}$

Get a result

$$J = \frac{4}{9} \epsilon_0 \left(\frac{2q}{m} \right)^{1/2} \frac{V_0^{3/2}}{d^2}$$

Child-Langmuir Law

⇒ spacecharge limits maximum achievable J for given V_0, d

Engineering form for electrons:

$$J = 2.33 \times 10^{-6} \frac{V_0^{3/2}}{d^2} \left[\text{A/m}^2 \right] \quad \begin{matrix} V_0 \Rightarrow \text{Volts} \\ d \Rightarrow \text{meters} \end{matrix}$$

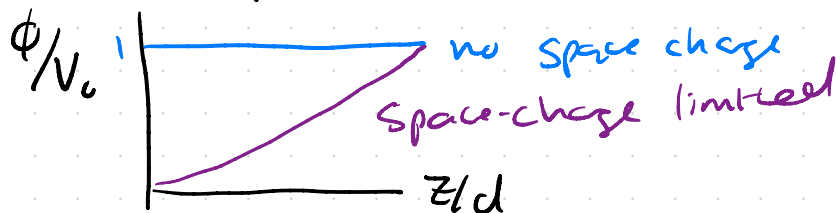
Limited by repulsion between charges

* CL limit is non-relativistic

Form of the potential:

$$\phi = V_0 \left(\frac{z}{d} \right)^{4/3} \Rightarrow E \text{ no longer uniform}$$

ϕ and E go to zero at cathode when space-charge limited



* slides

Electron emission

- Electron source: liberate bound electrons from some material (cathode)
- Three types of electron emission:

- Thermionic
- Photoemission
- Field emission

- Electrons are fermions, obey Fermi-Dirac statistics (recall fermions cannot occupy the same state)

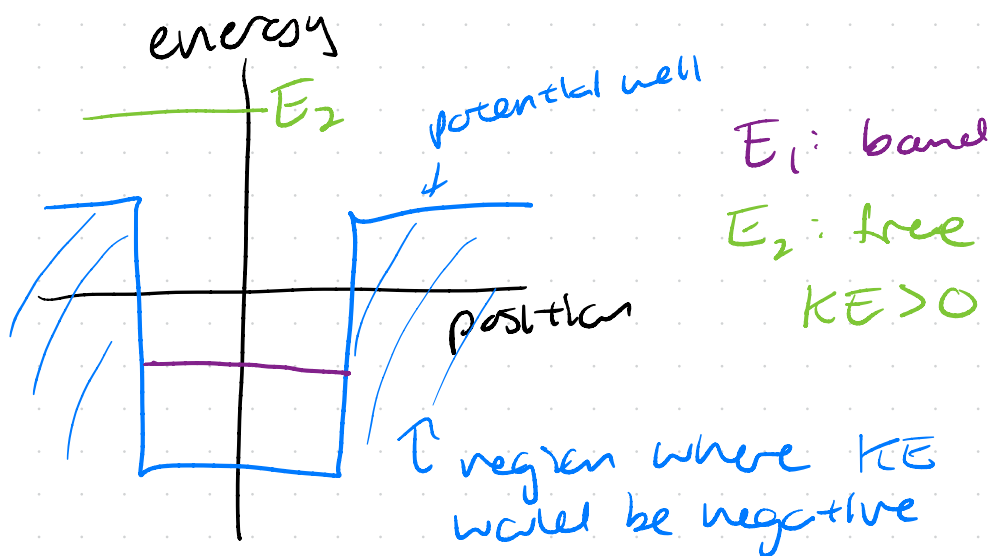
- At high temperatures, matches Maxwell-Boltzmann statistics * slide

$$f \rightarrow 1 \text{ for } E < E_F \text{ (occupancy)}$$

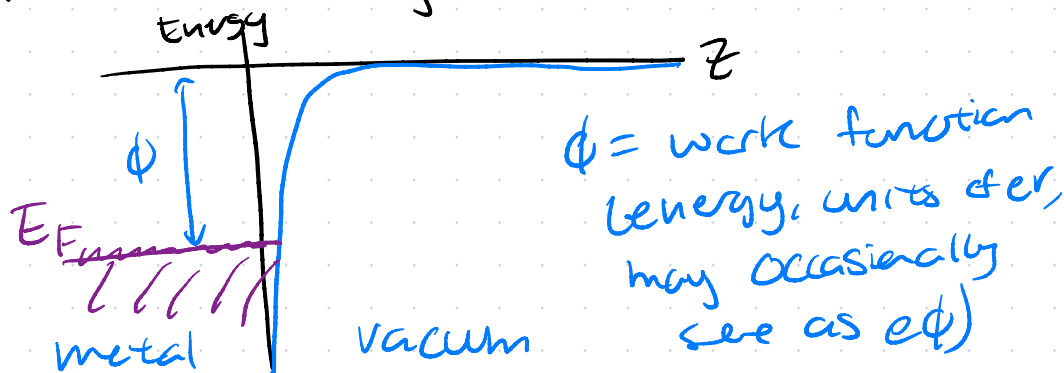
$$k_B = 8.617 \times 10^{-5} \text{ eV/K}$$

- Conventionally consider $V=0$ to be at level electron would not be bound

* quick review: $E = U + KE$



- Potential energy well

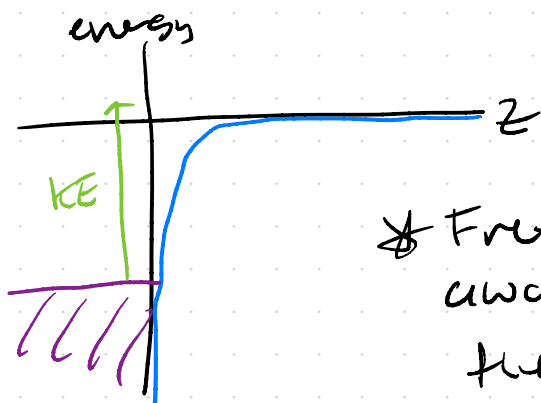


- Work function is a property of the material, $\sim eV$ for good conductors
- At non-zero T (temp) there is some energy spread around the Fermi level
 $\sim 10^{-2} eV$ for room temp

Thermionic emission

- Heating cathode material to overcome barrier
- with no external field, need

$$\frac{mv^2}{2} > \phi_{\text{work}} \Rightarrow v > \sqrt{\frac{2\phi_{\text{work}}}{m}}$$



* Free e^- accelerated away by applied field

- Recall: $E_S \approx 1.005 E_F$, MB statistics are a good approximation
- For some material with # density of electrons n_0 and work function ϕ

$$J_{\text{thermionic}} = 2n_0 e \left(\frac{2k_B T}{m} \right)^2 e^{-\phi_{\text{work}}/k_B T}$$

Increasing $T \rightarrow$ increasing current

* note: this term assumes not space charge limited

- Can describe normalized thermionic emittance with assumptions

↳ initial spread in transverse velocity due to temp, gives rise to transverse divergence as beam is emitted

- RMS beam size for x'

$$\sigma_{x'} = \frac{\langle p_x \rangle}{p_{\text{tot}}} = \frac{1}{\beta\gamma} \frac{\sqrt{\langle v_x^2 \rangle}}{c}$$

* We call: $\epsilon_{\text{rms}} \equiv \sqrt{\sigma_u^2 \sigma_{u'}^2 - \sigma_{uu'}^2}$

Assuming - no cross-correlation $\sigma_{uu'}$
- "laminar" flow of electrons, trajectories don't cross + current density is uniform

$$\epsilon_{\text{rms},x} = \sigma_x \sigma_{x'} \times \text{RMS emittance}$$

- we also defined normalized emittance which is invariant under acceleration:

$$\epsilon_N = \beta \gamma \epsilon$$

relative β and γ

$$\Rightarrow \epsilon_N = \beta \gamma \sigma_x \sigma_{x'} \Rightarrow \epsilon_N = \sigma_x \frac{\sqrt{\langle v_x^2 \rangle}}{c}$$

RMS velocity for MB distribution:

$$\langle v_x^2 \rangle = \frac{\int_0^\infty v_x^2 \exp(-mv_x^2/2k_B T) dv_x}{\int_0^\infty \exp(-mv_x^2/2k_B T) dv_x} = \frac{k_B T}{m}$$

$$\epsilon_{\text{thermionic}} = \sigma_x \sqrt{\frac{k_B T}{m c^2}}$$

higher T
higher ϵ

- Inherent emittance of beam based on its emitter

↳ sets the initial transverse phase space area, minimum phase space area of the beam

Photoemission

- Similar in basic idea (increase particle energy to exceed ϕ_{work}), but more complex process ~~8~~ slides

↳ Fundamentally: photo-electric effect

↳ Spicer 3 step model, transport through the bulk

- BULK absorption of light (not just surface):

Intensity at depth x :

$$I(x, \underbrace{h\nu}_{\substack{E = h\nu \text{ photon energy}}}) = I_0(h\nu) [1 - R(h\nu)] e^{-\alpha(h\nu)x}$$

$I_0(h\nu)$ initial intensity $\alpha(h\nu)$ absorption coefficient

$R(h\nu)$ reflectivity from surface

*note all energy-dependent $\rightarrow$ depends on energy levels in material; photocathodes can be metal, semiconductors

* Example laser energies UV
 $\sim 4.6 \text{ eV}$ $\Phi_{\text{work}} \Rightarrow \lambda = 270 \text{ nm}$
 $(E = h\nu, c = \lambda \nu)$

Some e^- excited from the bulk,
 Some from the surface

- Needs to be traveling towards the surface
- Needs to reach surface with enough energy to escape

$$\frac{P_z^2}{2m} > E_F + \Phi_{\text{work}} - \Phi_{\text{Schottky}}$$

reduction from $\uparrow$ surface electric field

* see Spicer paper for full derivation

Quantum Yield: # of e^- emitted per absorbed γ

$$QY(h\nu) = \frac{\frac{\alpha_{PE}}{\alpha} P_E}{1 + 1/\alpha L}$$

escape length $\uparrow$

P_E : prob. of excited e^- reaching surface with enough energy to escape

α_{PE} : Fraction of absorption which produces e^- with possibility of escape

* Also called quantum efficiency QE

• Major R&D area: optimizing QE in materials, construction

• Again have a range of $P_x \Rightarrow x'$, sets ε (see Douell notes)

$$\varepsilon_{\text{photo}} = \sigma_x \sqrt{\frac{h\nu - \phi_{\text{eff}}}{3mc^2}}$$

$\uparrow h\nu$

$\uparrow P_x \Rightarrow \uparrow x'$

$\Rightarrow \uparrow \varepsilon$

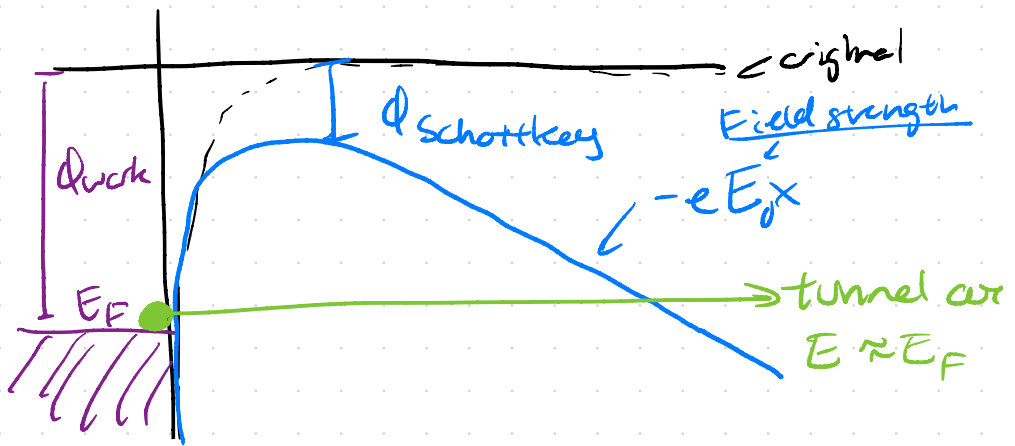
$$\phi_{\text{eff}} = \phi_{\text{work}} - \phi_{\text{Schottkey}}$$

Field emission

• Now: e^- not given any extra initial energy; instead, reduce barrier to allow tunneling

↳ Schottkey effect

• "cold" field emission $\Rightarrow$ no additional heating



- Depends on supply function of e^- vs energy, vs "barrier transparency" vs energy

$$J_{\text{field}} = \int n(E, T) D(E, E_0) dE$$

$\uparrow$ Temp $\uparrow$ Field strength
 $\uparrow$ electron energy

x see Danell

↳ determines energy spread

↳ depends on material, applied field

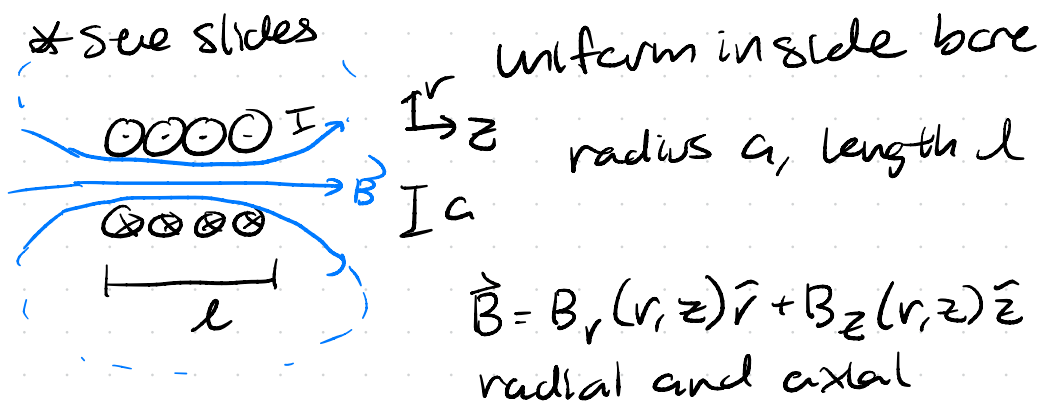
- Semi-empirical description:

Fowler-Nordheim eq (see slides)

- Not a nice expression for engineering

Solenoid focusing

* see slides



* see uploaded paper; some in HW CH6 and CH20 but not much

$$B_z(r, z) = B(z) - \frac{r^2}{4} B''(z) + \dots \quad B' \text{ etc gradients}$$

$$B_r(r, z) = -\frac{r}{2} B'(z) + \frac{r^3}{16} B'''(z) + \dots$$

$B_r \Rightarrow$ source of focusing

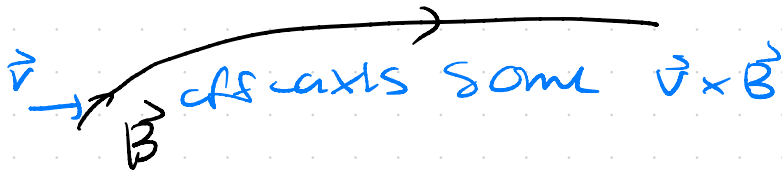
$$B_r(r, z) \approx -\frac{r}{2} B'(z) \quad \text{from gradient}$$

• Difference between this and quads:

axisymmetric field, focuses in both transverse planes simultaneously

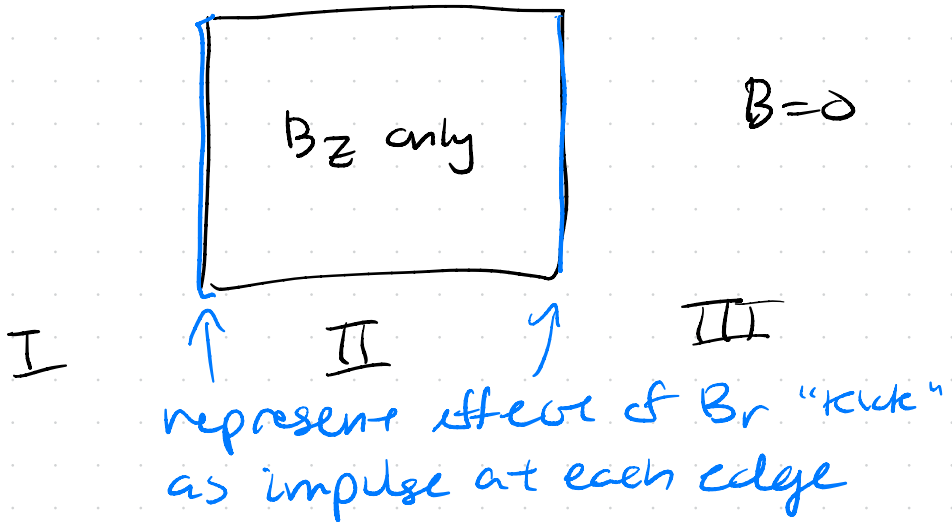


$\vec{v}$ on axis no effect



- Paper + building some conceptual understanding: use proximal approximation

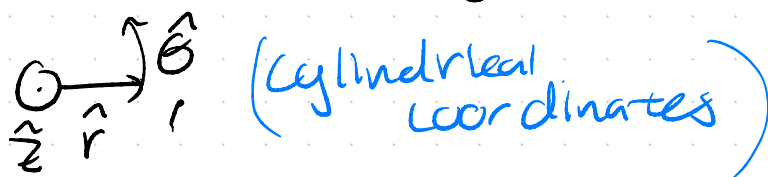
Consider 3 regions



$$B_r = -\frac{r}{2} B_0 [\delta(z) - \delta(z-L)] \quad \delta = \text{delta function}$$

- Same as "kick" previously, have kick in angle
- Start with: $\vec{V} = V_z \hat{z}$

Direction of force: $\hat{z} \times (-\hat{r}) = -\hat{\theta}$
azimuthal kick



$$\vec{F} = q \vec{v} \times \vec{B} = -q v_z B_r \hat{\theta} \Rightarrow \Delta V_{\theta} = r_0 \frac{q B_0}{2 \gamma m}$$

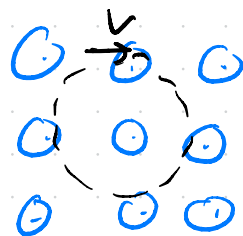
↳ kick depends on initial relativistic γ distance from center r_0

* small reduction in v_z , no change in total p

- Cyclotron frequency: $\omega_c = \frac{q B_0}{\gamma m}$
as we had before:

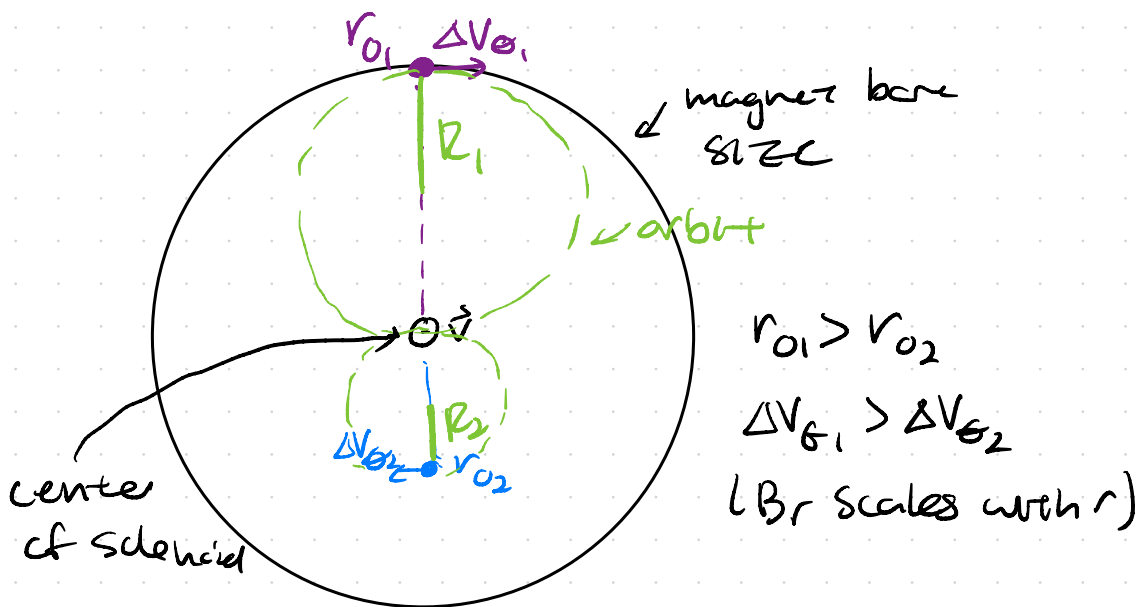
circular motion for $\vec{v} \perp \vec{B}$

$$\Rightarrow v_{\theta} \perp B$$



- Undergoes this circular motion around an off-axis center set by r_0

$$R = \frac{\gamma M V_{\theta}}{q B_0} = \frac{r_0}{2}$$



- Over time V_{θ} is not purely in $\hat{\theta}$
 - ↳ center of solenoid is at $r=0$
 - ✗ see slide

- At exit of magnet: gradient is in some other direction:

$\Rightarrow \Delta V_\theta$ impulse is in opposite direction

$$\vec{V} = V_r \hat{r} + V_\theta \hat{\theta} \quad \text{Force from Br}$$

- Circular motion is canceled, some non-zero V_r remains

$\hookrightarrow$ length has to be chosen correctly

- Can express this as a transfer M

Remember:
$$\begin{bmatrix} x \\ x' \end{bmatrix} = M(s|s_0) M \begin{bmatrix} x_0 \\ x'_0 \end{bmatrix}$$

* H.W. 20.1.2-20.1.3 coupled motion CH

- Solenoid focusing strength

$$K_{sol} = \frac{1}{4} S^2 (L_{sol})$$

Find from fields, solve EOM

* slide

length of solenoid

⇒ focal length

$$\frac{1}{f_{sol}} = \frac{1}{4} S^2 L_{sol} = \frac{1}{4} \left(\frac{q}{P} \right)^2 B_{sol}^2 L_{sol}$$

$\searrow$
 $\frac{1}{B\rho}$

very similar form
as for quadrupoles!

Thin lens:
 $f \gg L_{sol}$

Thin lens transfer matrix

$$M_{sol} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{f_{sol}} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{f_{sol}} & 1 \end{bmatrix}$$

Focusing in both planes

- Because this is based on edge effects, weak focusing + best for low energy beams like in a particle source

Iran Sources

& see slides

Thurs